

Preliminary mechanical design report for a simple robotic dog

Prepared by: Shaimaa Fayez Al-Safyani

University ID: 44251431

University major : Computer Engineering

Training provider : Smart Methods

Supervisor: Wessam Munshi

Introduction:

Designing a prototype of a simple robotic dog focusing on the fundamentals of mechanical design such as (frame, head, legs, joints, and motors).

Design Description:

- A frame was designed for the robot's body, and a head and four legs were added. Each leg has a joint and a motor to maintain the robot's stability.
- Number of joints: 4
- Degrees of freedom (DOF): Approximately 1 degree of movement (4 DOF)

Motor Selection:

Servo motors were chosen due to their ease of controlling the angle of joint rotation.

Calculating the initial torque of a joint:

- Given:

* Leg mass: 0.5 kg

* Leg length: 0.1 m

* Gravitational acceleration: 9.81 m/s^2

- Formula: Torque = Force $\times$ Distance

- Force: $F = 0.5 \times 9.81 = 4.905 \text{ N}$

- Torque: $T = 4.905 \times 0.1 \approx 0.49 \text{ N}\cdot\text{m}$

Stability and Center of Gravity:

The legs were distributed evenly across the body. To maintain the center of gravity and help the robot remain stable during movement.

Proposed Walking Method:

An alternating walking method was proposed, where the right front leg moves simultaneously with the left hind leg, then the left front leg moves simultaneously with the right hind leg to help balance the robot.

Expected Mechanical Problems:

- Instability of the robot during movement.
- Imbalance of body parts, leading to the robot falling.
- Weak motors, resulting in the robot being unable to support its own weight.

Conclusion:

A preliminary mechanical model of a robotic dog was designed, illustrating the basic design principles for standing and movement.

Final Design:



